9.8 A 9.3nV/rtHz 20b 40MS/s 94.2dB DR Signal-Chain Friendly Precision SAR Converter

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The precision Nyquist ADC space (>16b) is typically limited to throughput levels up to 15MS/s, the resolution of the converters >5MS/s being limited to 18b resolution [1-8]. Two-stage SAR ADCs dominate this space, but as the throughput increases the input and reference become difficult to drive, suffer from settling errors, and tend to degrade the FOM. This paper presents a two-stage SAR ADC with dynamically interleaved first stage and easy-to-drive input and reference signals, achieving 20b linearity and 9.3nV/rtHz noise spectral density, implemented in a 40nm CMOS process.

The kT/C noise is the dominant limiting factor of the speed limit above 16b resolution, as it requires a large sampling capacitor DAC. As the settling on the sampling capacitor requires ~14 time constants for 20b accuracy, the bandwidth of the driver and the ADC input filter must increase significantly. This results in excessive driver noise being sampled by the ADC and higher driver power consumption, as the driver equivalent noise bandwidth can be more than 15 times greater than the Nyquist frequency [1,4,5]. Time interleaving, typically used to increase converter sample rate, can also be leveraged for relaxing the constraints on the ADC driver and improving residue amplifier efficiency.

Figure 9.8.1 shows the architecture of the ADC. Two main differences are observable with respect to prior art [1-6]: the quantizer is fully separated from the residue generating DAC (RDAC), and the RDAC is composed of 8 slice DACs (sDAC) converting the signal, and 8 sDACs acquiring the signal, dynamically selected from a pool of 18 sDACs. Two spare sDACs are introduced to allow shuffling of the sampling DACs and spread the residual interleaving tones. Two phase interleaving of the RDAC extends the acquisition time to the full clock period, reducing the required sampling bandwidth by a factor of 2-4 times. Furthermore, the analog domain interleaving facilitates the residue amplifier (RA) efficiency optimization. An integrating RA is not well-suited due to offset, 1/f noise and incomplete reference settling (at 20b level) during the RA phase. At this conversion rate and precision level, power-cycling the RA is not feasible, leading to ~50% idle time. Multiplexing of the RA stage, which otherwise would be the dominant power consumer, significantly improves the efficiency.

A key goal is to make driving the input of the ADC easy. Using a single quantizer would disturb the input after the completion of the conversion (~7ns), compromising the critical RDAC input sampling. Therefore, a second quantizer has been introduced, despite the FOM penalty. The flash-ADC can be shared, thanks to a <1ns return to acquisition time.

To maximize the SNR, the design makes use of 3.3V transistors throughout the first-stage ADC in order to allow a $6V_{pp}$ input range. The 3.3V MOSFETS come at channel length cost (400nm) and significant speed degradation. To overcome the speed limitation, the quantizers are built out of one slice DAC. The matched-quantizer sDAC with the RDAC allows >11b matching, resulting in 9b quantization in the first stage. The RDAC works as a follower of the quantizer with the conversion results loaded as they develop. Several benefits arise from this. First, the reference current is minimal (1.8mA) as the DAC is always set directly to the right state, being comparable to that of a 15MS/s converter [5]. Second, the RDAC reference switches, which are the dominant distortion contributor due to nonlinear capacitance connecting to ADC input, can be made smaller, thereby reducing the distortion. Third, the reference switches can be toggled with slower edge rates thereby reducing the high-frequency glitching of the reference which otherwise results in ringing. Fourth, the RDAC can be segmented to a higher level as the binary-to-unary decoder timing is not critical. Fifth, the 1.1V to 3.3V level shifters can be smaller and more power efficient. Sixth, a dynamic-element-matching algorithm can be used in conjunction with dither, as the shuffler is not in the critical timing path.

The obvious drawback of the interleaving comes from the offset, gain, timing and bandwidth mismatch artefacts of the DACs. The offset and gain error are typically easier to resolve as they are signal-frequency independent, to a first degree. The bandwidth mismatch is resolved by increasing the DAC sampling bandwidth to more than 10× the Nyquist frequency. The timing mismatch is minimized by collocating the top plate, sampling, switches and clocking them with the same signal enabled via a series NFET (Fig. 9.8.2). Aside from the RDAC, the sampling switch array includes the switches for the quantizer sDAC. The critical sampling edge (falling) of the sampling switches is set by the TP clock signal, while the return to acquisition edge (non-critical) is marked by the rising edge of the enable clock. A hold-low FET is introduced to keep the gate of the sampling switch low once the enable signal transitions low and the start of the conversion phase commences. This allows the TP_clock signal to be shared between channels and return high, marking the acquisition of the complementary set of 8 sampling DACs. The remaining timing mismatch is dominated by the V_T mismatch of the sampling NFET, but this is significantly improved by placing the devices in the same line of diffusion and by dummying. The estimated residual RMS timing mismatch is ~0.6ps. The usage of 1.1V NFET for the sampling switches, required for fast sampling edges, would not be possible without the 3b flash ADC, as the top plate signal can reach ±3V during the MSB trial.

Another key point of the architecture is that the input signal and the RA residue are sampled at the same time, requiring a single quiet time instance during the conversion period. This is a key signal-chain feature in achieving the 20b performance, as the crosstalk requirement from the interface and other interferers is significantly relaxed.

Despite the reduced reference current resulting from the usage of a separate quantizer, the RDAC would still draw approx. 1.8mA reference current. This is too high to be provided from an external reference voltage: e.g. a 0.1Ω resistor would produce signal and throughput-dependent errors of approx. 30LSB. An internal reference buffer would introduce 1/f noise, offset drift with temperature and would need to have very good load regulation. This is resolved by employing a dynamic reference buffer (Fig. 9.8.3 - top), which takes advantage of the RDAC not requiring an accurate reference voltage during the conversion phase. Thus, a coarse pre-charge reference buffer is used to source the charge required by the RDAC during the SAR bit trials. Before starting the RA phase, the reference multiplexer is toggled to the accurate input voltage reference, REF_IN, settling only residual errors. Due to the reference incomplete settling at the beginning of the RA phase, the residue is only ~15b accurate. Complete (20b) settling of the reference, takes approx. 65% of the RA phase. Figure 9.8.3 (top) shows the residual reference current. Linearization of the DAC using dynamic element matching (DEM) is always a challenge in high-speed converters, due to the time penalty. Outstanding linearization of the DAC transfer function has been demonstrated in lower throughput implementations [6]. Despite the DEM randomization not being in the critical timing path, thanks to the separate quantizer and RDAC, the time budget for this is limited to 1ns. A simple 3b round-robin shuffler achieves the target linearity within the time constraint. Additionally, the linearity is limited by other, DAC-extrinsic, nonlinearity errors such as: top-plate switch leakage, reference settling and interface cross-talk. One advantage of the analog-interleaved architecture is that each of the 8 sDACs composing the RDAC can be considered a unary DAC segment. The overall DEM range becomes two-dimensional, with 6b total range. Figure 9.8.3 (bottom) shows a random selection of the DAC slices with two degrees of freedom for any unit capacitor. Dither is used to complement the DEM linearization of the DAC. The architecture allows the dither range of the quantizer to be different from that of the RDAC. Thus, the dither range is enhanced by 2b when the result is transferred from the quantizer to the RDAC.

The ADC is implemented in 40nm CMOS technology with a die area of 2.25mm² and ADC core area of 0.9mm². The conversion rate is 40MS/s, with ±3V differential input range. The typical measured INL is +2.2/-1.8ppm and the DNL is ±0.27ppm, extending the monotonicity well above 20b. The DAC weight correction is done on-chip with coefficients derived externally, suitable due to very good DAC stability. The offset stability is 0.05ppm/°C while the gain stability is 0.32ppm/°C. The input current is linear with $350\mu A_{PP}$ magnitude, allowing low-power drivers, normally used for sub-10MS/s ADCs. The ADC input filter bandwidth is reduced by a factor of ~2.5× due to full-period input signal acquisition. The resulting signal chain FOM improvement is ~3-4dB. The noise floor is flat across the Nyquist range without any distinguishable interleaving spurs. The noise spectral density is 9.3nV/rtHz. The chip power consumption at 40MS/s is 69mW with full-scale sinewave, and the ADC core power is 50mW. The dynamic range is 94.2dB and SNR <100kHz is 93.7dB including external driver noise (~0.3dB). At 1MHz THD is 107dB, HD2=-115dB and HD3=-109dB, with the SNR=92dB (affected by signal source noise). The Schreier FOM is FOM_{S_DR}=180.2dB and FOM_{S_SNR}=179.7dB.

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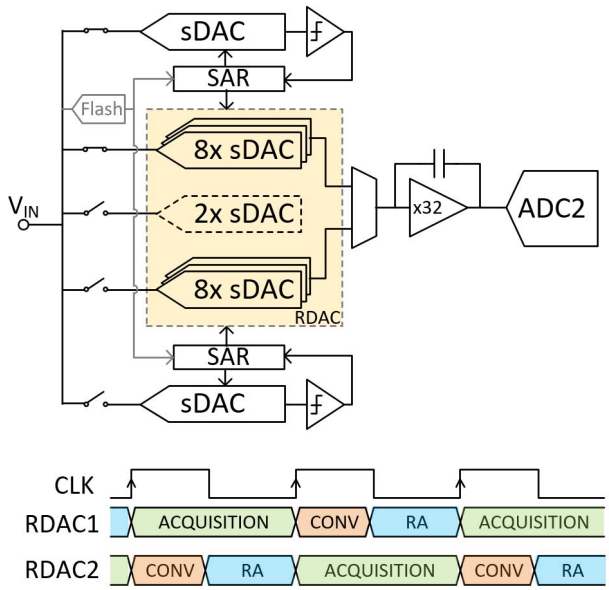


Figure 9.8.1: Front-end of the analog-interleaved ADC architecture.

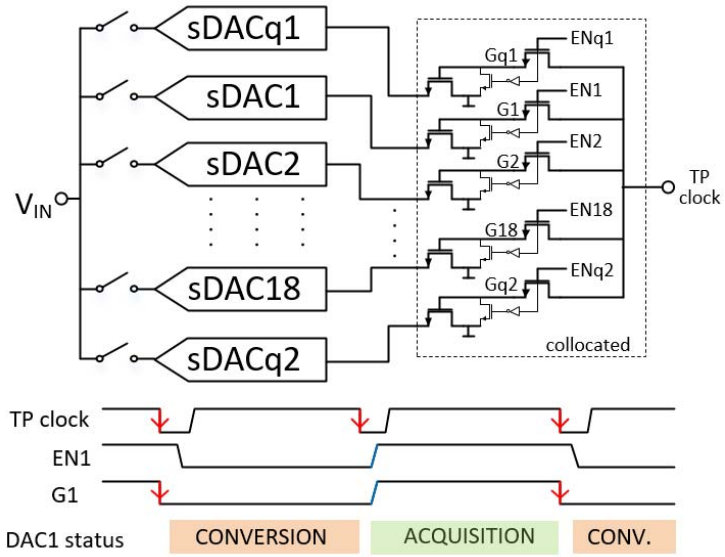


Figure 9.8.2: DAC top plate sampling circuitry.

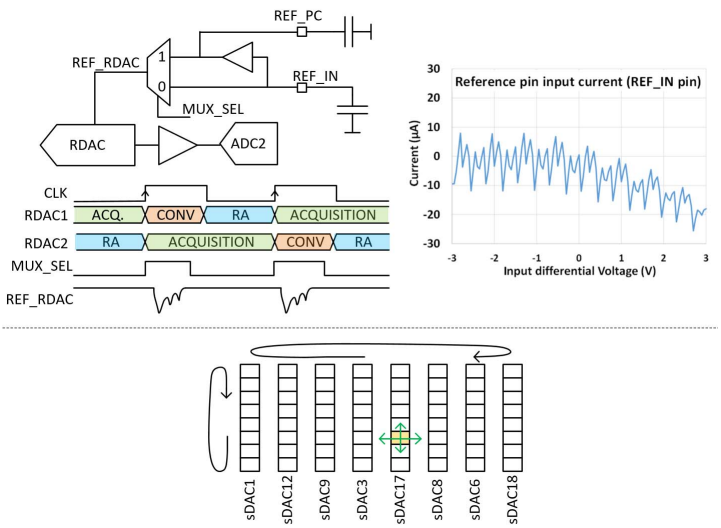


Figure 9.8.3: Pre-charge reference buffer architecture and input reference current (top); two-dimensional shuffling of the unary DAC capacitors (bottom).

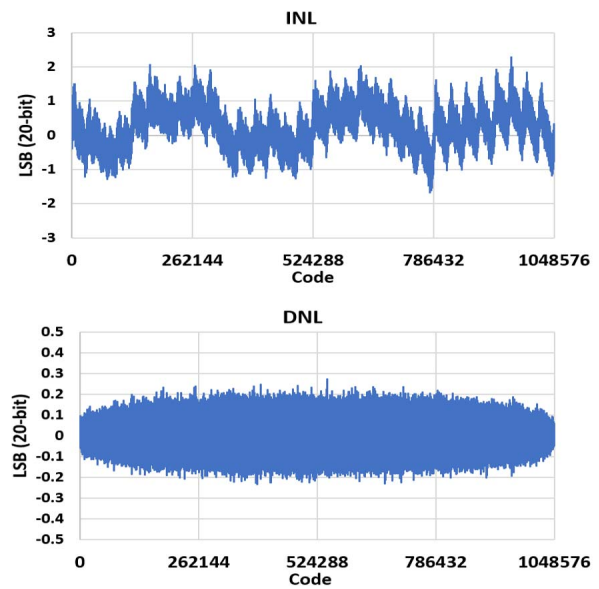


Figure 9.8.4: INL and DNL (20b level).

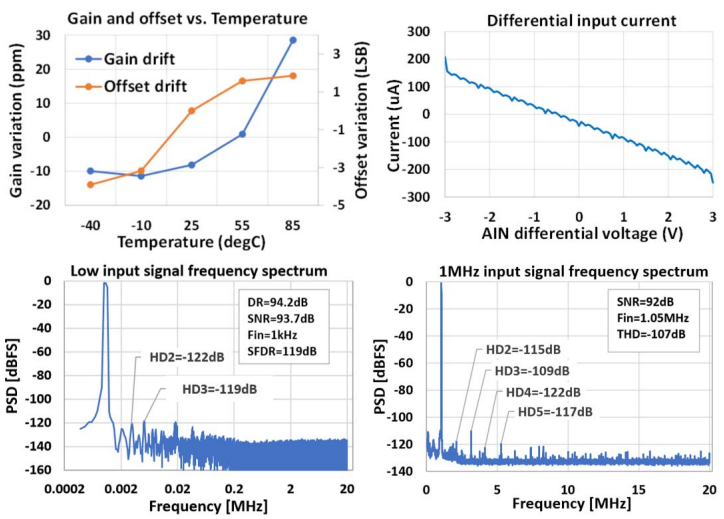


Figure 9.8.5: Measured static and dynamic performance.

	This work	Hurrell [1]	ElShater [2]	Li [3]	Bannon [4]	Steens-gaard [6]	Shen [8]	LTC2387-18 [5]	ADC3583 [7]
Process [nm]	40	250	180	180	180	180	55	-	-
F_S [MHz]	40	12.5	15	12	5	2	16	15	65
Resolution [bit]	20	18	16	16	18	24	16	18	18
Noise density [nV/rtHz]	9.3	26	17.2	18.8	21.9	17.1	103.9	15.9	11.8
INL [ppm]	±2.2	±32	±16	±24	±2	0.03	±36.8	±2.4	±28
SNR/DR [dB]	93.7/94.2	91/93	92.4/93.9	-/94	99/100.2	106/106.3	78/-	96/-	84.5/-
SFDR [dB]	119	93	100.7	100.8	-	-	98	119	95
Power core [mW]	50	105	9.8	9.43	30.5	8.5	16.3	-	-
Power system* [mW]	69	-	-	-	-	15	-	125	122
FOM_{S_DR} core [dB]	180.2	170.7	182.7	182.1	179.3	187.0	-	-	-
FOM_{S_DR} system* [dB]	178.8	-	-	-	-	184.5	-	-	-
FOM_{S_SNR} core [dB]	179.7	168.7	181.2	-	178.1	186.7	165	-	-
FOM_{S_SNR} system* [dB]	178.3	-	-	-	-	184.2	-	173.8	168.8

System power includes reference buffers, internal LDOs, monitoring and protection, etc and excludes the interface.
FOM_{S_DR}=DR+10log(BW/Power) FOM_{S_SNR}=SNR+10log(BW/Power)

Figure 9.8.6: Performance summary and comparison to representative >16b converters.

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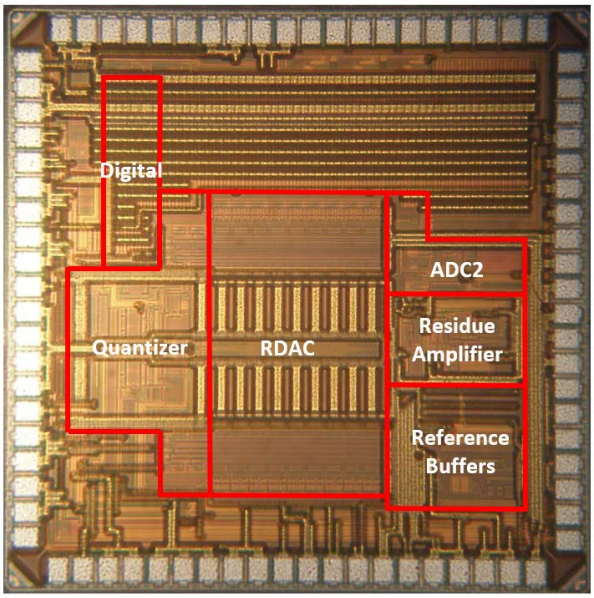


Figure 9.8.7: Die micrograph.

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